Quiz Bowl RL Buzzer

Learning when to buzz in multiple-choice pyramidal quiz bowl

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Poster uses a concise subset of the report plus visuals from the slide deck.

github.com/hass0114/qanta-buzzer

Problem + Setup

- Quiz bowl reveals clues incrementally, so the agent must decide when to act, not only what to answer.
- We study a controlled multiple-choice version to isolate the stopping decision and make evaluation reproducible.
- At each prefix the policy chooses WAIT or BUZZ(answer) under rewards that trade correctness against timing.

Why this setting works

- Open-ended QA mixes buzzing with aliasing and grading noise. MC keeps the gold answer in the set and focuses the task.
- Distractors are screened for alias collisions, question overlap, and easy surface artifacts.
- We report system score, buzz accuracy, timing, and control tests instead of raw accuracy alone.

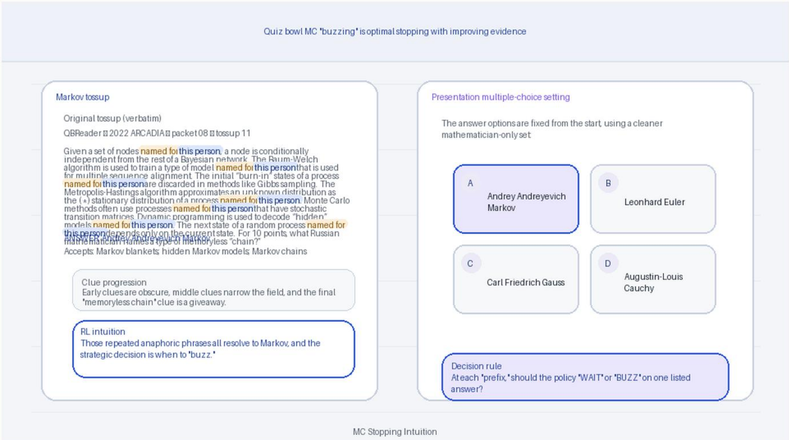
POMDP view

Observation	clue prefix h_t + answer set C
Action	WAIT or BUZZ(i), i in {A, B, C, D}
Policy	p(WAIT) = 1 - p_buzz; p(BUZZ i) = p_buzz * p_ans(i)
Goal	stop when committing beats waiting

Contributions

- An MC quiz-bowl environment that turns buzzing into a clean sequential decision problem.
- A factored policy view: one module answers, another decides when the evidence is strong enough to buzz.
- Control evaluations that test whether the agent uses clue content instead of answer-set artifacts.

Running example

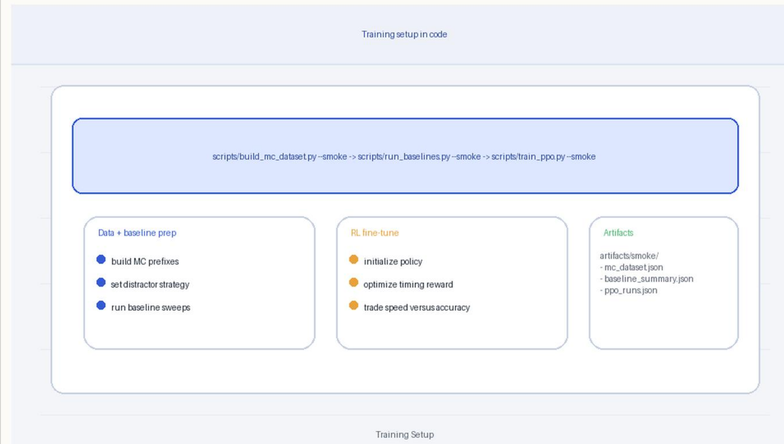


The Markov tossup shows how repeated cues collapse the posterior toward one answer.

Method

- Build MC prefixes from QANTA tossups and score each option with profile-based likelihood models.
- Convert beliefs into structured observations: posterior mass, entropy, step index, and similarity features.
- Warm-start with supervised answer learning, then fine-tune a stopping policy with PPO.

Training pipeline



Data prep, baseline sweeps, and PPO fine-tuning are implemented as separate stages.

Decision dynamics

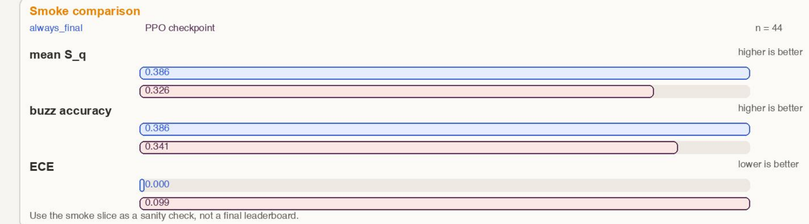
As more clues arrive, the posterior sharpens and the stop decision becomes easier.

Optimal rule: buzz once the value of committing exceeds the value of waiting.

Experiments + Results

- Task: given a clue prefix and four answer options, choose WAIT or BUZZ(answer).
- Smoke evaluation snapshot: best baseline is always_final, while the PPO checkpoint tests whether RL changes timing behavior.

Smoke comparison



Use the smoke slice as a sanity check, not a final leaderboard.

Analysis + controls

- Choices-only accuracy is 9.1%, below the 25% chance level.
- Shuffle leaves mean S_q at 0.237, so content still matters after clue reordering.
- Alias substitution produces mean S_q 0.243; this path is still a limited control because alias lookup coverage is sparse.

Conclusions

- The project reframes quiz-bowl buzzing as an optimal stopping problem with explicit uncertainty.
- The factored policy gives a clean way to separate answering from timing.
- Current smoke full-eval snapshot: mean S_q 0.243, buzz accuracy 38.6%, mean step 3.50.

References

- Rodriguez et al. (2019) QANTA and incremental QA.
- Schulman et al. (2017) Proximal Policy Optimization.
- Boyd-Graber and Daume (2012/2013) Bayesian quiz-bowl buzzing.